DS4420 Project - Indoor Scene Recognition

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Introduction/Motivation

Indoor scene recognition is **essential for service systems that help with accessibility and safety**, especially for the visually impaired/blind.

Reliable GPS signals are often not available, which is why navigation systems on autonomous robots rely on visual recognition. For a robot to preform a task, like drive to the closest chair in the living room, it must first recognize what part of its environment is the living room. These systems can eventually become something that is reliably able to guide the visually impaired through an unknown environemnt.

The current issues surrounding Indoor Scene Recognition is that **it is extremely expensive to train your own system**. They are also very heavy, often requiring a server to do computations. Since current indoor recognizers are trained to recognize objects in the space and making a classification based on these objects, there is a big class imbalance, as there will be more nightstands than fire extinguishers for example.

Taking into consideration of these issues, we wanted to create our own system that addressed theses issues. We wanted something that was **lightweight** but also was **robust against class imbalances** from the data.

Dataset

We used the MIT Indoor Scene Recognition database. This database includes around 15,000 images along with their annotations of objects in the picture. Refer to Figure 1 for the distribution

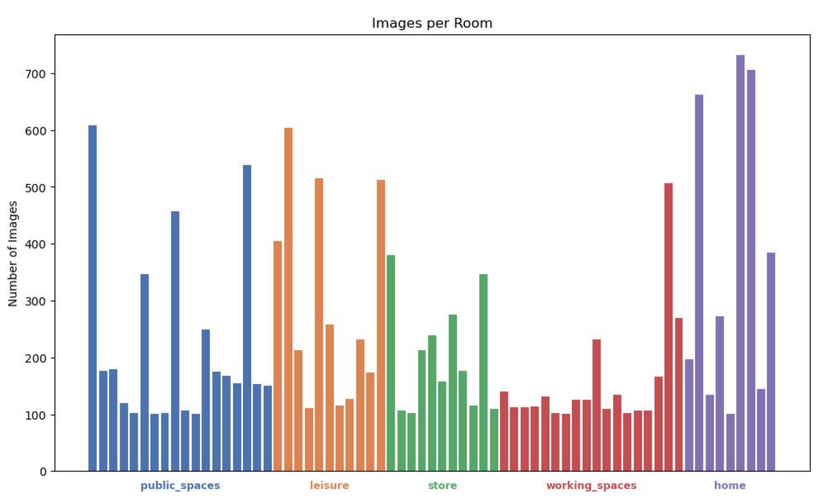

Figure 1: Image count for each room, for each category

Methodology

Data Preprocessing

- We split our image data, originally ~70 categories, into 5 main categories: stores, homes, public spaces, leisure and working spaces
- We turned the images to grayscale then into numpy arrays ready for the CNN process.

CNN (Python)

- We implemented a CNN model with 3 convolutional layers (0.25 dropout each) -> 2 hidden ReLU layers (0.5 dropout each) -> softmax output, trained on 20 epochs.
  - Layers were tuned so our model doesn't overfit, but still maintains a reasonable accuracy

Bayesian Modelling (R)

- We extracted the 128 feature embeddings per category, then used those features as our input features for Bayesian logistic regression
  - Bayesian Metropolis-Hasting sampler was manually implemented, since logistic likelihood has no closed-form posterior
  - Normalized CNN features to match our isotropic Gaussian prior
  - Thinned samples and kept every 50th out of 10000 samples

Results

- Our CNN model ended with a 50.5% accuracy with a softmax output layer, which is ~30% better than randomly guessing (1/5 chance)
- Our Bayesian Model with the CNN features recorded a final accuracy of around 62.9% accuracy, which means our Bayesian model did successfully improve our CNN model quite a bit
- In terms of individual categories, our model was best at predicting a home with ~75% accuracy, and worst at predicting a public space at ~53.54% accuracy
- However our chains showed high autocorrelation and slow mixing even after thinning, suggesting that sampler could still be improved

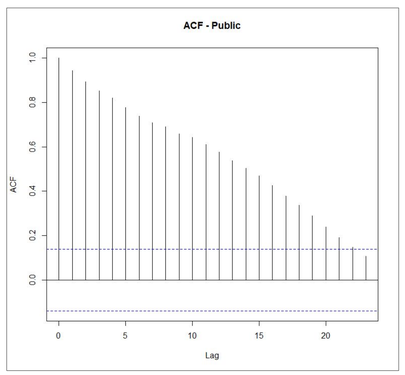

Figure 2: ACF Plot for Public Spaces Category

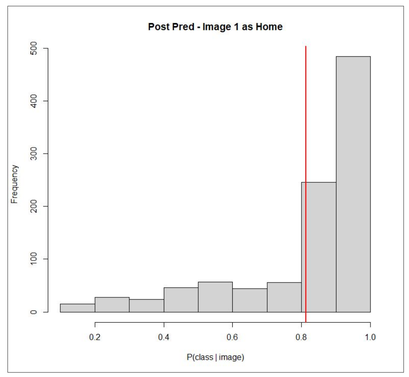

Figure 3: Posterior Distribution of Image 1 in Test Set

Conclusion

- Our Bayesian model built on CNN features did end up having an accuracy of 62.5%, which is pretty good considering the scope of our project
  - Due to constraints in programs and costs, we had to scale to 128x128 grayscale images
  - CNN isn't complex enough for object detection, which as a human is one of the main ways to differentiate between rooms/scenes
- However we couldn't solve the high autocorrelation in our manual MCMC chains even after thinning, most likely due to our feature space having 128 features, and our output being between 5 categories, which makes it hard to implement manually
- Overall, our project of combining Bayesian logistic regression with CNN was relatively successful, we showed that a Bayesian approach that gives uncertainty distributions was somewhat valuable, especially for scene recognition where prediction confidence matters also matters

Future Work

In the future, if we ever get more computation power, we would love to train more complex models that can predict more specific categories, and see if the bayesian modelling in conjunction with CNN feature extraction, can make an accurate predictor for Indoor Scene Recongition.

We would also like to work with an even bigger dataset, while 15,000 images is a lot, each room type typically have 100-200 images, which is almost impossible to train a model that converges

References

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